Formal Verification

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1 Introduction to the Computational Certificate

The complexity of the 13-manifold construction necessitates a high-degree of verification beyond manual derivation. To this end, we provide a formal constructive certificate developed in the `Lean 4` interactive theorem prover. This approach ensures that the prime-counting sieve logic, which serves as the arithmetic foundation for the manifold's boundary conditions, is logically sound and machine-verified. By providing a computational witness where $\pi(100) = 25$, we establish the empirical validity of the sieve within the formalized 13-manifold architecture.

1.1 Structure of verification_code_instructions Folder

The following components are included:

1. **verification_code** The Sub-Folder.
2. `readme_instructions.txt`: A streamlined guide for the reviewer, detailing the steps to initialize the Lean server and verify the certificate goals.
3. **Lean_Logic_Guide.pdf**: Serves as the formal logic exposition for "Recursive Rounding: A Constructive Proof of the Riemann Hypothesis."
4. **Verification.pdf**: This document.

1.2 Structure of the verification_code Sub-Folder

The following components are included:

1. `sieve.lean`: The core formalization script. It contains the inductive definitions of the manifold structures and the terminal evaluation of the prime-counting function.
2. `lean-toolchain`: A version-lock file specifying `Lean 4 v4.27.0`. This ensures that the code executes in an environment identical to the one used during the proof's development.
3. `lakefile.lean`: The project configuration file used by the `Lake` build system to manage dependencies and compilation targets.
4. `lake-manifest.json`: Version-lock manifest ensuring exact reproducibility of all project dependencies.
5. `README.md`: Project overview and architectural map for the 13-manifold formalization.

2 Setup

The certificate is authored in **Lean 4.27.0**.